Methods & Q&A prep — Spin-polarized stellarator NWL

Internal notes (not for presentation)

Thaddaeus Kiker

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Roadmap

Part A — methods & concepts, one per slide, explained from paper + code.

Part B — anticipated PI questions, with grounded answers.

Honesty line to keep straight all meeting: the *analytic* result is cross-checked against an exact free-streaming quadrature to machine precision; the *OpenMC* Monte-Carlo pipeline on Ginsburg is in bring-up — machinery and geometry validated, the full MC-vs-exact-vs-analytic *numbers* comparison on a real high-$\varepsilon_{\mathrm{eff}}$ equilibrium is the *next* milestone, not yet done.

Part A

Methods & concepts

A1. Quadrature, and my hybrid quadrature

Quadrature = approximating a definite integral by a weighted sum at chosen nodes, $\int f \approx \sum_i w_i f(\phi_i)$ (here Gauss–Legendre in the toroidal angle ϕ).

- **Two ways to do the per-loop ϕ-integral** $g = \int \mathcal{J}\mathcal{N}\, K/D^{3/2}\, \mathrm{d}\phi$: (i) reduce it to closed form (elliptic F, E + elementary) via the series; (ii) just evaluate it numerically.
- **My hybrid hybridizes the two:** keep the integrand *exact and analytic* (no ε-truncation) — $\mathcal{J}, \mathcal{N}, D, \hat{\boldsymbol{B}}$ as explicit differentiable functions of the design parameters — but do the single residual ϕ-quadrature *numerically*.
- **Why:** it has *no* $\varepsilon_{\mathrm{eff}}$ *ceiling* below grazing and, unlike a black-box numerical NWL, *retains exact autodiff gradients* in all design parameters. This is Lion's cost regime but differentiable.

A2. What I did to Schwartz's method for QA plasmas

Schwartz (2025) is *exact* for axisymmetric rings: a planar circle makes $D_0(\phi) = p^2 + r^2 + z^2 - 2pr\cos\phi$ affine in $\cos\phi$, so $w = e^{i\phi}$ puts the integral on a genus-1 (elliptic) curve $\Rightarrow$ closed form in F, E.

- A stellarator loop is non-planar; the period integral becomes **hyperelliptic** (genus ≥ 2) — no finite closed form in F, E.
- **My addition:** a *field-period perturbation series*. Split each loop into mean $+\,\varepsilon$ ripple, expand D^{-s}, and every term lands in the canonical form (trig polynomial)$/D_0^{(2j+1)/2}$ — reducible to the *same* F, E.
- Leading order $g^{(0)}$ *is* Schwartz's kernel verbatim; corrections are additive ripple terms layered on the existing Carlson-form code. Reduces *exactly* to the axisymmetric kernel at zero ripple (regression test, 2×10^{-16}).
- Controlling parameter is the *geometric* non-axisymmetry ε_{eff}, not $|B|$ quasisymmetry — QA (gently shaped) is where the truncated series converges fastest.

A3. The spin-polarized DT source in OpenMC

A self-contained `CompiledSource` (`src/polarized_fusion_source.cpp`, `src/spf_sampler.hpp`), no private MIT code.

- **Anisotropic emission:** monoenergetic 14.1 MeV; birth direction sampled from $I(\theta) = \frac{3}{4}a\sin^2\theta + (\frac{2}{3}b + \frac{1}{3}c)(\frac{1}{4} + \frac{3}{4}\cos^2\theta)$ by *stratified* sampling of the two angular shapes (exact inverse-CDF, cross-checked vs rejection). Rate factor $\eta = a + \frac{2}{3}b + \frac{1}{3}c$ kept *separate* from the direction PDF.
- **Quantization axis = local field:** $\cos\theta = u \cdot \hat{\boldsymbol{B}}(\boldsymbol{r}_s)$ with $\hat{\boldsymbol{B}}$ from `field_->bhat()`; sample in a $\hat{\boldsymbol{B}}$-aligned frame, rotate to global with a Gram–Schmidt basis (no gimbal degeneracy).
- **Plugin loading:** OpenMC loads the compiled `.so` via `CompiledSource(library=...)`; thread-safe (K_{pol} state set at construction, all sampling `const`, RNG only `openmc::prn(seed)`).
- Verified: A/iso $= \frac{3}{2}\sin^2\theta$ to 0.3%/bin, mean 1.0000; B≡C shape; linearity of mixes.

A4. Analytic work in the free-streaming limit

- **Reduction engine:** harmonic recursion $\mathcal{C}^{(s)}_{k+1} = 2h\mathcal{C}^{(s)}_{k} - \mathcal{C}^{(s)}_{k-1} - 2\mathcal{C}^{(s-1)}_{k}$, master recursion for $J_a = \int (h - \cos\phi)^a \, \mathrm{d}\phi$, F/E seeds through the existing Carlson R_F/R_D layer. *No third-kind (Π) integral ever appears.*
- **Polarized kernels:** $\mathrm{A} = \mathrm{iso} - \cos^2$, $\mathrm{B/C} = \frac{1}{4}\mathrm{iso} + \frac{3}{4}\cos^2$; one extra power $1/D$ raises the elliptic order (s: $5/2 \to 7/2 \to 9/2$). Order-by-order cross-check $g^{(j)}_A + g^{(j)}_c = g^{(j)}_{\mathrm{iso}}$ from $\sin^2 + \cos^2 = 1$.
- **Gradient closure:** $\partial_h \mathcal{C}^{(s)}_{k} = -s\,\mathcal{C}^{(s+1)}_{k}$ — differentiation maps the integral family into itself, so optimization gradients cost *no* new special-function calls and give a second, independent generator of the elliptic tower.
- **Robustness:** singularity subtraction for grazing; Leibniz boundary term for the moving visible arc; multi-arc visibility ($\mathcal{N} = \hat{\boldsymbol{n}} \cdot \boldsymbol{\Delta} > 0$).

A5. Elliptic integrals, and where they enter

Incomplete elliptic integrals of the first / second kind ($m = k^2$):

$$F(\varphi, m) = \int_0^{\varphi} \frac{\mathrm{d}x}{\sqrt{1 - m\sin^2 x}}, \qquad E(\varphi, m) = \int_0^{\varphi} \sqrt{1 - m\sin^2 x}\,\mathrm{d}x$$

- They are the closed form for $\int (h - \cos\phi)^{-1/2}\,\mathrm{d}\phi$ and $\int (h - \cos\phi)^{+1/2}\,\mathrm{d}\phi$ — the half-integer-power integrals the flux factor ($D^{-3/2}$) and polarization kernels ($D^{-5/2}$) produce.
- Seeds: $J_{-1/2} \propto F(\phi_\pm/2, m)$, $J_{+1/2} \propto E(\phi_\pm/2, m)$, modulus $m = -2/(h-1) = -4pr/((p-r)^2 + z^2)$, $h = (p^2 + r^2 + z^2)/2pr$.
- Implemented through **Carlson symmetric forms** R_F, R_D — numerically stable, differentiable, and they handle the *asymmetric* visible arcs (evaluate at $\phi_+/2$ and $\phi_-/2$, difference) without modification.

A6. How the perturbation expansion was set up

Split loop into mean + ripple (formal order parameter ε):

$$R(\phi) = p + \varepsilon\, dR(\phi), \qquad Z(\phi) - z_w = z + \varepsilon\, dZ(\phi)$$

dR, dZ expanded in field-period harmonics $\cos / \sin(nN_{\mathrm{fp}}\phi)$. Then *exactly*:

$$D = D_0 + \varepsilon D_1 + \varepsilon^2 D_2, \quad D_1 = 2[dR(p - r\cos\phi) + z\, dZ], \quad D_2 = dR^2 + dZ^2$$

- Jacobian $\mathcal{J}$, flux numerator $\mathcal{N} = \mathcal{N}_0 + \varepsilon\mathcal{N}_1$, and field projection $\hat{\boldsymbol{B}} \cdot \boldsymbol{\Delta} = P_0 + \varepsilon P_1 + \varepsilon^2 P_2$ expand the same way.
- Insert into g, use $D^{-s} = D_0^{-s}(1 + \varepsilon D_1/D_0 + \varepsilon^2 D_2/D_0)^{-s}$, collect powers of $\varepsilon \Rightarrow$ strict series $g = g^{(0)} + \varepsilon g^{(1)} + \varepsilon^2 g^{(2)} + \cdots$.
- **Implementation:** represent $dR, dZ, \mathcal{N}, P, D_1, D_2, \ldots$ as Fourier coefficient vectors; form products by *convolution*; contract against $\mathcal{C}_k^{(s)}, \mathcal{S}_k^{(s)}$. General; avoids hand-derived coefficients.
- Verified $g^{(1)}, g^{(2)}$ vs finite differences of the exact integral to 9–11 sig figs at *asymmetric* limits.

A6b. The subtle second-order polarized term

Naive expansion $\hat{\boldsymbol{B}} \approx \hat{\boldsymbol{B}}_0 + \varepsilon\tilde{\boldsymbol{b}}$ gives $P_2 = \tilde{\boldsymbol{b}} \cdot \boldsymbol{\Delta}_1$. But $|\hat{\boldsymbol{B}}| = 1$ is only enforced to *first* order by $\hat{\boldsymbol{B}}_0 \cdot \tilde{\boldsymbol{b}} = 0$.

- At second order the unit-norm constraint demands a field term $\varepsilon^2\tilde{\boldsymbol{b}}_2$ with $\hat{\boldsymbol{B}}_0 \cdot \tilde{\boldsymbol{b}}_2 = -\frac{1}{2}|\tilde{\boldsymbol{b}}|^2$, contributing $\tilde{\boldsymbol{b}}_2 \cdot \boldsymbol{\Delta}_0$ to P_2.
- This is the second-order Taylor remainder of the angled-field numerator P_0 in the field-angle excursions α_1, β_1 (closed-form trig polynomials).
- **Why it matters:** omit it and the A, B/C modes converge only at *second* order; include it and they recover the *third*-order convergence of the isotropic series. Small for QA (small $|\tilde{\boldsymbol{b}}|$), but it is the correct closed form. Verified to 9 sig figs.

A7. What the hybrid solution is

Three regimes, separated by two thresholds; the hybrid bridges them:

- $\varepsilon_{\text{eff}} \lesssim \frac{1}{2}$: **truncated series** — fast, closed form, differentiable (2 terms $\to$1% for $\varepsilon_{\text{eff}} \lesssim 0.15$; 3 terms $\to\approx 0.3$).
- $\frac{1}{2} \lesssim \varepsilon_{\text{eff}} \lesssim \varepsilon_{\text{graze}} \approx 2$: **hybrid quadrature** — exact integrand, single numerical ϕ-integral, exact gradients retained. Reproduces ground truth to $\sim 2 \times 10^{-9}$.
- Near grazing ($\min_\phi |\boldsymbol{\Delta}| \to 0$): **singularity-subtracted hybrid** — closed-form peak model removed, fixed node count suffices ($\sim$12 nodes).
- True contact / scattered load: **Monte Carlo** required.

The series is the *analytic backbone* that makes every stage differentiable; the hybrid is the same integrand, not a different model.

A8. Other concepts the work leans on

- **Boozer coordinates / quasisymmetry:** QA/QH are statements that $|B|$ depends on Boozer angles only through a fixed combination — they govern *confinement*, not directly this expansion.
- ε_{eff} **vs qs-error:** my controlling parameter is $\varepsilon_{\text{eff}} \sim \max_\phi \sqrt{dR^2 + dZ^2} / \min_\phi |\mathbf{\Delta}_0|$ — *real-space* boundary shaping over standoff. The quasisymmetry error (deviation of $|B|$ from exact QS) is a *different* number; a QA config can be excellent in $|B|$ yet strongly shaped geometrically.
- **TBR:** tritium breeding ratio — needs a breeder blanket + scattering; the free-streaming inboard/outboard current split is a geometric *precursor*, not a TBR.
- **Spin-state fractions** (a, b, c)**:** from deuteron/triton spin projections; $a{+}b{+}c{=}1$; nonpolarized $= (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ gives $\sigma = \frac{2}{3}\sigma_0$. B and C share angular shape (differ only in rate).

Part B

Anticipated PI questions

B1. Is the derivation right, and physically valid?

Derivation. Every piece is verified numerically: seeds $J_{\pm 1/2}$ (2×10^{-15}), master recursion (6×10^{-14}), harmonic recursion (9×10^{-12}), h-derivative closures vs autodiff ($\sim 10^{-14}$), $g^{(0)}, g_c^{(0)}$ vs exact quadrature ($\sim 4 \times 10^{-16}$), $g^{(1)}, g^{(2)}, g_c^{(1)}, g_c^{(2)}$ vs finite differences (limited only by step). The Appendix-C worked single-harmonic example reproduces exactly.

Validity. It is the *free-streaming / first-flight* (uncollided) wall load: no scattering, no multiplication. That is a genuine physical limit (the uncollided tally of any transport code), *and* it is exactly why it is a clean ground truth. The assumption is ideal field-aligned polarization; a fraction < 1 interpolates toward isotropic and leaves the geometry untouched.

B2. Convergence and error control

- Truncation error at order p scales as $\varepsilon_{\text{eff}}^{p+1}$; fitted log–log slopes $1.00, 2.03, 3.01$ (iso) and $\approx$ same for A, B/C.
- Practical table (representative loop): $\varepsilon_{\text{eff}} = 0.10 \to 2.9\%/0.12\%/0.024\%$; $0.30 \to 7.1\%/1.6\%/0.72\%$; $0.60 \to 6.8\%/11\%/7.1\%$ (non-monotone — series has left its radius near $\varepsilon_{\text{eff}} \approx \frac{1}{2}$).
- **A-posteriori control:** the first neglected term $|g^{(p+1)}|$ is itself an error estimate, computable *without* the exact answer; it tracks the true error up to $\varepsilon_{\text{eff}} \approx 0.4$. Applied *patch by patch* — inboard loops that breach the bound are routed to the hybrid, closed form everywhere else.
- Two breakdowns at *separated* ε_{eff}: series radius $\approx \frac{1}{2}$ (expansion property) vs grazing ≈ 2 (geometry property).

B3. Why QA? And ε_{eff} vs quasisymmetry error

- **Why QA:** it is the regime geometrically *closest* to axisymmetric, so the perturbation about Schwartz's exact kernel converges fastest; and the polarized field-direction ripple $\tilde{\boldsymbol{b}}$ is milder than in QH, so the polarized series does not lag the isotropic one.
- **Key subtlety to state plainly:** ε_{eff} is set by real-space shaping, *not* by $|B|$ symmetry. The published QAs are strongly shaped *because* that is the price of quasisymmetric $|B|$: ESTELL 1.01/1.47, NCSX 1.27/1.78, ARIES-CS 1.57/1.73, WISTELL-A 1.55/2.39 (median/max ε_{eff} at a tight wall).
- **Consequence:** for real QAs at a tight standoff the truncated series alone is *not* sufficient over most of the wall — the differentiable hybrid is the operational tool; the series is the exact leading order + low-ε_{eff} cross-validation anchor.

B4. Validation status — verified vs in progress

Verified (analytic side):

- Every reduction/recursion/correction vs direct quadrature & finite differences to machine precision; ε^{p+1} law confirmed; zero-ripple regression to axisymmetric kernel (2×10^{-16}); fwd/rev autodiff agree (3×10^{-12}).
- Analytic series vs *exact free-streaming quadrature* on a QA config: $\sim$1% iso, few-% polarized at $\varepsilon_{\text{eff}} = 0.15$. Hybrid vs ground truth $\sim 10^{-9}$.
- Free-streaming OpenMC source vs analytic on low-ε_{eff} QA (toy $\varepsilon_{\text{eff}} = 0.08$ and *published* QUASR 59509 $\varepsilon_{\text{eff}} = 0.43$): $\leq 1.8\%$ on most walls; inboard 4–6% residual under strong shaping (bounded, not a bug).

In progress (MC side, honest): the OpenMC pipeline on Ginsburg is in *bring-up* — sampler, field-map reader, geometry, end-to-end machinery validated; the full *MC-vs-exact-vs-analytic numbers* comparison on a real high-ε_{eff} equilibrium (e.g. precise_QA) is the *next* milestone, *not yet done*. No claim that MC numbers are validated against the analytic theory.

B5. Novelty vs Schwartz / Lion / Bae

- **vs Schwartz (2025):** he is exact only in axisymmetry (planar ring $\rightarrow$ elliptic). I remove the hyperelliptic obstruction perturbatively, keeping closed form + differentiability in a *stellarator*; leading order is his kernel verbatim.
- **vs Lion (2022):** he does the toroidal direction *numerically* (black-box, no gradients). My hybrid matches his cost in the strongly-shaped limit but keeps *exact autodiff gradients* for shape/polarization optimization.
- **vs Bae (2025):** they show spin-polarization steering matters via Monte Carlo in spherical tokamaks. I give the *analytic, differentiable free-streaming* counterpart and extend the geometry to stellarators; their MC and my analytic are complementary (scattered vs uncollided).
- **New physics deliverable:** closed-form *polarized* A and B/C stellarator kernels with the $|\hat{\boldsymbol{B}}| = 1$ second-order term, and the ∂_h gradient-closure result.

B6. Spin-polarization physics — realism

- **Reactivity:** parallel-aligned (A) enhances the DT cross section ∼50% (σ_0 vs $\frac{2}{3}\sigma_0$ nonpolarized); the C combination suppresses it (to $\frac{1}{3}\sigma_0$). Reaction goes through the $J^\pi = \frac{3}{2}^+$ ^{5}He resonance.
- **Angular redistribution:** A mode radiates $\propto \sin^2\theta$ (perpendicular to $\hat{\boldsymbol{B}}$); B/C $\propto \frac{1}{4} + \frac{3}{4}\cos^2\theta$ (toward $\hat{\boldsymbol{B}}$). This is what makes the wall load a *functional of field geometry*.
- **Quantization axis:** the local field $\hat{\boldsymbol{B}}$ — spins precess rapidly about it, so polarization is referenced to $\hat{\boldsymbol{B}}$; both codes use the *local* $\hat{\boldsymbol{B}}$ at the source.
- **Realism caveats (honest):** I assume ideal field-aligned polarization, spatially uniform (a, b, c), no depolarization during confinement. Depolarization mechanisms (wall collisions, resonances, ∇B) set the achievable fraction and profile — not modeled; a fraction < 1 just interpolates toward isotropic.

B7. Geometry and DAGMC choices

- **Analytic side:** non-planar source loops; wall taken conformal to the boundary at a uniform poloidal gap; convex-cross-section visibility (single visible arc), with multi-arc ($\mathcal{N} > 0$) handling wired into the driver.
- **MC side now (honest, per DEFERRED.md):** the 3D physics is in the *source* (field map + flux-surface births); the transport *geometry* is still an axisymmetric layered box-torus, square cross-section — **no field-period geometry effect claimed yet**.
- **DAGMC** (Direct Accelerated Geometry Monte Carlo) lets OpenMC track on real CAD/triangulated 3D surfaces — needed to build a true 3D stellarator first wall from the boundary Fourier fits (via Stellarmesh / build123d). That is the upgrade path from the box-torus.
- DAGMC ships with OpenMC's **linux-64** build as an optional feature; unavailable on my local aarch64 / Mac — hence Ginsburg.

B8. Why HPC / Ginsburg?

- **DAGMC availability:** only on the linux-64 build — the cluster gives it; local arm/Mac does not.
- **Statistics:** resolved per-patch, per-mode, ϕ-resolved wall tallies across *many* configurations (QA→QH scan) need large history counts; deep-coil tallies need variance reduction (weight windows, MAGIC) — HPC scale.
- **The high-ε_{eff} regime:** real QAs sit at $\varepsilon_{\text{eff}} \sim 1$–$2.5$; there the analytic series diverges *and* even fixed-node exact quadrature can break down near grazing (precise_QA: ≈34% exact-vs-hybrid disagreement, unphysical negatives). That regime is **MC-only** — the science question lives there.
- Headline science it unlocks: how *quasisymmetry* controls the *directional benefit* of spin polarization at the wall, across configurations, with scattering included.

B9. Concrete next steps

1. Finish OpenMC bring-up on Ginsburg; run the **free-streaming** MC and close the three-way *numbers* cross-check (MC = exact quadrature = analytic) per patch/mode on a real equilibrium — the gated milestone.
2. Turn materials on: quantify the *scattering departure* from free-streaming (the part the analytic cannot give).
3. Build a real 3D wall via DAGMC (boundary Fourier → Stellarmesh) to claim a genuine field-period *geometry* effect.
4. **QA→QH quasisymmetry scan**: correlate the steering benefit / loading with (non-)axisymmetry across configs.
5. Joint *wall-shape* + *polarization* co-optimization with exact analytic gradients (the differentiability payoff); analytic-paper + MC-paper split.

B10. Traps to not fall into (self-reminders)

- Do *not* say the MC numbers are validated against the analytic theory — they are not yet; only machinery/geometry are.
- Do *not* conflate ε_{eff} (geometry) with quasisymmetry quality ($|B|$). QA $\neq$ geometrically axisymmetric.
- NWL here is free-streaming, normalized to constant fusion rate $\Rightarrow$ pure directionality; it is *not* a shielded dose, *not* a TBR.
- In a *fixed* square geometry both polarized modes can *raise* the global peaking factor; the B/C benefit is the *inboard center-stack* load reduction, not a lower global peak (that needs wall re-shaping).
- B and C share angular shape (a common notes error is calling C isotropic — it is not).